General Mathematics — Exercise Sheet 37

Local equations and critical points

Variant: Statements

Exercise 1 (Local coordinates and a choice of branch). Let $F(x, y) = (x + y, xy)$. Compute its Jacobian at $(2, 1)$ and use Lecture 37's cited inverse theorem to state the local conclusion there [C]. Then prove directly that the restriction to $V = \{(x, y) : x > y\}$ is a bijection onto $W = \{(s, p) : s^2 - 4p > 0\}$ with a C^1 inverse. Find the inverse and its Jacobian at $(s, p) = (3, 2)$. Explain why no neighbourhood of a point (c, c) admits an injective restriction of the full map. Use Lecture 8's square-root derivative and Lecture 21's matrix inversion. The direct branch verification does not need the cited existence theorem.

Exercise 2 (Two dependent coordinates and higher derivatives). Near $(t, x, y) = (0, 1, 0)$ consider the equations

$$x^2 + y^2 = 1 + t, \qquad xy = t.$$

Use Lecture 37's cited implicit theorem to obtain a unique local branch $(x(t), y(t))$ through $(1, 0)$ [C]. Find $x'(0)$ and $y'(0)$. Prove that the branch is C^2 by examining its first-derivative formula, then find $x''(0)$ and $y''(0)$ and the two quadratic Taylor expansions. Use Lectures 8 and 36's chain rules and Lecture 37's Taylor formula. All local branch claims and their derivative values inherit [C].

Exercise 3 (A degenerate Hessian with a parameter). For $a \in \mathbb{R}$, let $f_a(x, y) = x^4 + ax^2y^2 + y^4$. Show that the origin is critical with zero Hessian for every a. Classify it for every real parameter by analysing the function itself: state when it is a strict minimum, a nonstrict minimum or a saddle. Specify whether each minimum is global. Use Lecture 37's definitions and ordinary algebra, rather than treating zero determinant as a classification.

Exercise 4 (A Taylor polynomial and a certified error). Let $f(x, y) = \ln(1 + x + y) + \sin(x - y)$ on $1 + x + y > 0$. Find its quadratic Taylor polynomial at the origin. For $|x| + |y| \le 1/4$, prove the error bound

$$|f(x, y) - P_2(x, y)| \le \frac{155}{162}(|x| + |y|)^3.$$

Give the polynomial value and a rational error bound at $(1/20, -1/10)$. Explain why the restriction to the line $x = 0$ has a strict local maximum at zero, while the full two-variable function has no local extremum there. Use Lecture 37's Taylor/Fermat results and Lecture 11's finite scalar Taylor remainder.

Exercise 5 (A regular constraint and its extrema). Use the following necessary condition with its announced sketch status [S] and cited implicit-existence input [C]. If f, g are C^1 near $a \in \mathbb{R}^n$, $n \geq 2$, $g(a) = 0$ and $\nabla g(a) \neq 0$, a local extremum of f among nearby points satisfying $g = 0$ obeys $\nabla f(a) = \lambda \nabla g(a)$ for some $\lambda \in \mathbb{R}$. The number λ is a *Lagrange multiplier*. A sketch is as follows. Choose a coordinate with nonzero partial derivative of g and use Lecture 37's cited implicit theorem to make the constraint a local graph. Apply the scalar Fermat condition to f on that graph. The chain rule and implicit derivative formula make its gradient proportional to ∇g. This is a necessary condition, not an existence or classification assertion.

Apply it to $f(x, y, z) = 2x - y + 2z$ on $x^2 + 2y^2 + 4z^2 = 1$. Find all multiplier candidates. Prove directly, by Lecture 12's Cauchy–Schwarz inequality, that they are the unique global maximum and minimum. The final inequality verification is independent of the cited local existence theorem.

Exercise 6 (Least squares with nonconstant abscissas). Let $n \geq 2$, let $x_j, y_j \in \mathbb{R}$ for $1 \leq j \leq n$, and suppose the x_j are not all equal. Find the unique global minimum over $(\alpha, \beta) \in \mathbb{R}^2$ of

$$S(\alpha, \beta) = \sum_{j=1}^{n} (y_j - \alpha - \beta x_j)^2.$$

Prove uniqueness, not just the normal equations. Apply the result to the data

$$(0, 1), \quad (1, 2), \quad (2, 2), \quad (3, 4).$$

Finally explain the change when all $x_j = c$ for one real c. Use Lectures 21 and 37's linear-system and critical-point results and ordinary finite-sum algebra.